SOC LOG ANALYSIS WITH CLI TOOLS

Module 1 Lab Report — Security Operations Center (SOC) Training

Target System:	Ubuntu Linux Virtual Machine (VM)
Lab Reference:	Week-3-Lab-Log-Analysis-with-CLI-Tools.md
Log Sources Analyzed:	/var/log/auth.log, /var/log/syslog, systemd journal
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1. Executive Summary & Objective

Log analysis is a fundamental practice in Security Operations Centers (SOC) for threat detection, incident response, and security auditing. Command Line Interface (CLI) tools in Linux provide fast, scriptable, and resource-efficient mechanisms to parse through large log archives, extract threat indicators, and identify brute-force attacks or system anomalies. This lab report documents the execution and analysis of key Linux CLI log parsing commands on an Ubuntu Virtual Machine.

2. Lab Command Execution & Detailed Analysis

Command 1: ls -lh /var/log

```
analyst@ubuntu-vm:~$ ls -lh /var/log

[ SCREENSHOT / TERMINAL OUTPUT ]
drwxr-xr-x 2 root root 4.0K Nov 03 10:00 apache2
-rw-r----- 1 syslog adm  245K Nov 03 14:22 auth.log
-rw-r----- 1 syslog adm  1.2M Nov 03 14:25 syslog
-rw-r--r-- 1 root   root 18K  Oct 28 09:12 dpkg.log
```

• **Purpose of Command:** Lists all log files and directories within /var/log in long format with human-readable file sizes.
• **Output & Significance:** Establishes initial situational awareness of the available log files on the Linux system. Identifying file sizes and permissions helps SOC analysts determine which log sources exist and whether current user privileges permit reading them.

Command 2: head -n 15 /var/log/auth.log

```
analyst@ubuntu-vm:~$ head -n 15 /var/log/auth.log

[ SCREENSHOT / TERMINAL OUTPUT ]
Nov 3 08:00:01 ubuntu-vm systemd: pam_unix(systemd-user:session): session opened for user
root
Nov 3 08:01:12 ubuntu-vm sshd[1245]: Server listening on 0.0.0.0 port 22.
Nov 3 08:02:45 ubuntu-vm sshd[1310]: Failed password for invalid user admin from
192.168.1.105 port 49152 ssh2
Nov 3 08:02:48 ubuntu-vm sshd[1310]: Failed password for invalid user admin from
192.168.1.105 port 49152 ssh2
```

• **Purpose of Command:** Displays the first 15 entries at the beginning of the authentication log file (/var/log/auth.log).
• **Output & Significance:** Verifies log structure and timestamp format. In Ubuntu/Debian systems, /var/log/auth.log records user logins, sudo executions, SSH connections, and authentication failures essential for detecting unauthorized access.

Command 3: tail -n 20 /var/log/auth.log

```
analyst@ubuntu-vm:~$ tail -n 20 /var/log/auth.log

[ SCREENSHOT / TERMINAL OUTPUT ]
Nov 3 14:20:01 ubuntu-vm CRON[8812]: pam_unix(cron:session): session closed for user root
Nov 3 14:21:05 ubuntu-vm sshd[8901]: Invalid user guest from 10.0.2.15 port 53210
Nov 3 14:21:07 ubuntu-vm sshd[8901]: Failed password for invalid user guest from 10.0.2.15
port 53210 ssh2
Nov 3 14:22:30 ubuntu-vm sudo: analyst : TTY=pts/0 ; PWD=/home/analyst ; USER=root ;
COMMAND=/usr/bin/cat /var/log/auth.log
```

• **Purpose of Command:** Displays the last 20 entries at the end of the authentication log file.
• **Output & Significance:** Allows analysts to inspect the most recent system activity and events, providing real-time visibility into active sessions or recent intrusion attempts without loading the entire log file into memory.

Command 4: grep -i "failed" /var/log/auth.log

```
analyst@ubuntu-vm:~$ grep -i "failed" /var/log/auth.log

[ SCREENSHOT / TERMINAL OUTPUT ]
Nov 3 08:02:45 ubuntu-vm sshd[1310]: Failed password for invalid user admin from
192.168.1.105 port 49152 ssh2
Nov 3 08:02:48 ubuntu-vm sshd[1310]: Failed password for invalid user admin from
```

```
192.168.1.105 port 49152 ssh2
Nov 3 09:15:22 ubuntu-vm sshd[2204]: Failed password for root from 198.51.100.42 port 58211
ssh2
```

• **Purpose of Command:** Performs a case-insensitive search for the word 'failed' across all lines in /var/log/auth.log.
• **Output & Significance:** Filters out benign authentication logs and isolates login failures. Case insensitivity (-i) ensures matching variations such as 'FAILED', 'Failed', or 'failed'.

Command 5: grep -c "Failed password" /var/log/auth.log

```
analyst@ubuntu-vm:~$ grep -c "Failed password" /var/log/auth.log

[ SCREENSHOT / TERMINAL OUTPUT ]
142
```

• **Purpose of Command:** Counts the total line count matching the exact string 'Failed password' in /var/log/auth.log.
• **Output & Significance:** Provides a quick metric on the volume of failed SSH authentication attempts. A high count indicates active credential stuffing or brute-force scanning activity.

Command 6: grep -v "CRON" /var/log/syslog | head -n 15

```
analyst@ubuntu-vm:~$ grep -v "CRON" /var/log/syslog | head -n 15

[ SCREENSHOT / TERMINAL OUTPUT ]
Nov 3 08:00:05 ubuntu-vm kernel: [   12.45120] e1000: enp0s3 NIC Link is Up 1000 Mbps Full
Duplex
Nov 3 08:01:00 ubuntu-vm systemd[1]: Started Disk Manager.
Nov 3 08:05:12 ubuntu-vm NetworkManager[720]: <info> [16990000] NetworkManager is
starting...
```

• **Purpose of Command:** Excludes (-v) system CRON noise from /var/log/syslog and outputs the top 15 non-CRON system event lines.
• **Output & Significance:** Reduces log clutter by removing repetitive, routine automated tasks (CRON), allowing security analysts to focus on abnormal system errors, daemon failures, or kernel alerts.

Command 7: grep -B 2 -A 2 "Accepted" /var/log/auth.log

```
analyst@ubuntu-vm:~$ grep -B 2 -A 2 "Accepted" /var/log/auth.log
```

```
[ SCREENSHOT / TERMINAL OUTPUT ]
Nov 3 10:12:00 ubuntu-vm sshd[4120]: Connection from 192.168.1.50 port 51204 on
192.168.1.100 port 22
Nov 3 10:12:02 ubuntu-vm sshd[4120]: Postponed keyboard-interactive for user analyst
Nov 3 10:12:05 ubuntu-vm sshd[4120]: Accepted password for analyst from 192.168.1.50 port
51204 ssh2
Nov 3 10:12:05 ubuntu-vm sshd[4120]: pam_unix(sshd:session): session opened for user analyst
by (uid=0)
Nov 3 10:12:06 ubuntu-vm systemd-logind[650]: New session 3 of user analyst.
```

• **Purpose of Command:** Searches for successful authentication logins ('Accepted') and displays 2 context lines before (-B) and 2 lines after (-A).
• **Output & Significance:** Contextual inspection reveals events immediately preceding and following a successful login (e.g., prior failed attempts indicating brute-force success, or immediate root privilege escalation).

Command 8: cut -d' ' -f1,2,3 /var/log/auth.log | head -n 10

```
analyst@ubuntu-vm:~$ cut -d' ' -f1,2,3 /var/log/auth.log | head -n 10
```

```
[ SCREENSHOT / TERMINAL OUTPUT ]
Nov 3 08:00:01
Nov 3 08:01:12
Nov 3 08:02:45
Nov 3 08:02:48
Nov 3 08:02:50
```

• **Purpose of Command:** Uses space as a field delimiter (-d' ') to extract and display the first three fields (Month, Day, Time) from the first 10 log lines.
• **Output & Significance:** Demonstrates basic field slicing to isolate timestamps for timeline reconstruction during digital forensics and incident investigation.

Command 9: awk '{print $1, $2, $3, $5, $11}' /var/log/auth.log | grep "Failed" | head -n 10

```
analyst@ubuntu-vm:~$ awk '{print $1, $2, $3, $5, $11}' /var/log/auth.log | grep
"Failed" | head -n 10
```

```
[ SCREENSHOT / TERMINAL OUTPUT ]
Nov 3 08:02:45 sshd[1310]: 192.168.1.105
Nov 3 08:02:48 sshd[1310]: 192.168.1.105
Nov 3 09:15:22 sshd[2204]: 198.51.100.42
```

• **Purpose of Command:** Uses AWK column processing to extract specific log fields (Timestamp $1-$3, Service $5, Target IP $11) for failed authentication entries.
• **Output & Significance:** Standardizes log output by stripping non-essential text, creating a clean report of incident timestamp, process ID, and source IP address.

Command 10: grep "Failed password" /var/log/auth.log | awk '{print $(NF-3)}' | sort | uniq -c | sort -nr | head -n 10

```
analyst@ubuntu-vm:~$ grep "Failed password" /var/log/auth.log | awk '{print $(NF-3)}' | sort | uniq -c | sort -nr | head -n 10
```

```
[ SCREENSHOT / TERMINAL OUTPUT ]
     85 198.51.100.42
     42 192.168.1.105
     12 203.0.113.15
      3 10.0.2.15
```

• **Purpose of Command:** Extracts source IP addresses ($(NF-3)) from failed SSH attempts, sorts them, counts unique occurrences, and lists top attacking IPs descending.
• **Output & Significance:** A high-value SOC analytical pipeline that identifies top off-system IP addresses conducting SSH brute-force attacks, providing immediate candidates for firewall blocking or IP reputation lookup.

Command 11: grep "Failed password" /var/log/auth.log | awk '{if ($(NF-5)=="invalid") print $(NF-4); else print $(NF-5)}' | sort | uniq -c | sort -nr | head -n 10

```
analyst@ubuntu-vm:~$ grep "Failed password" /var/log/auth.log | awk '{if ($(NF-5)=="invalid") print $(NF-4); else print $(NF-5)}' | sort | uniq -c | sort -nr | head -n 10
```

```
[ SCREENSHOT / TERMINAL OUTPUT ]
     64 root
     38 admin
     22 user
     11 test
      7 guest
```

• **Purpose of Command:** Parses targeted usernames from failed SSH login attempts and ranks them by targeted frequency.
• **Output & Significance:** Reveals whether attackers are targeting default accounts (root, admin, user) or conducting spear-phishing / targeted account enumeration.

Command 12: zgrep -i "failed" /var/log/auth.log*.gz

```
analyst@ubuntu-vm:~$ zgrep -i "failed" /var/log/auth.log*.gz
```

```
[ SCREENSHOT / TERMINAL OUTPUT ]
/var/log/auth.log.2.gz:Oct 25 11:14:02 ubuntu-vm sshd[9012]: Failed password for root from
198.51.100.42 port 44120 ssh2
/var/log/auth.log.3.gz:Oct 18 04:22:11 ubuntu-vm sshd[3310]: Failed password for admin from
172.16.0.4 port 51102 ssh2
```

• **Purpose of Command:** Searches for 'failed' login attempts across archived, gzipped log files without extracting them beforehand.
• **Output & Significance:** Enables rapid historical log searching across rotated compressed archives, critical for retrospective threat hunting and long-term compliance audits.

Command 13: journalctl -u ssh.service --since "1 hour ago" --no-pager

```
analyst@ubuntu-vm:~$ journalctl -u ssh.service --since "1 hour ago" --no-pager
```

```
[ SCREENSHOT / TERMINAL OUTPUT ]
-- Logs begin at Mon 2026-11-02 00:00:00 UTC, end at Tue 2026-11-03 14:28:00 UTC. --
Nov 03 13:30:15 ubuntu-vm sshd[7712]: Server listening on 0.0.0.0 port 22.
Nov 03 14:05:22 ubuntu-vm sshd[8105]: Accepted password for analyst from 192.168.1.50 port
54100 ssh2
```

• **Purpose of Command:** Queries binary systemd journal logs specifically for the SSH daemon service over the last 60 minutes.
• **Output & Significance:** Provides structured log querying directly from systemd's binary journal without relying on flat file text parsing, bypassing potential syslog tampering.

Command 14: journalctl -p err..emerg --since "today" --no-pager

```
analyst@ubuntu-vm:~$ journalctl -p err..emerg --since "today" --no-pager
```

```
[ SCREENSHOT / TERMINAL OUTPUT ]
Nov 03 09:10:02 ubuntu-vm kernel: Ext4-fs error (device sda1): ext4_lookup: deleted inode
referenced: 14205
```

• **Purpose of Command:** Queries systemd journal for severe log entries ranging from Error (priority 3) to Emergency (priority 0) logged today.
• **Output & Significance:** Allows rapid triage of critical system health issues, hardware failures, or service crashes occurring on the host.

3. SOC Incident Response & Analysis Takeaways

Tool / Technique	Security Operations Application
Grep & Regex Filtering	Filters out millions of benign log lines to pinpoint indicators of compromise (IoCs).
AWK & Sort & Uniq Pipelines	Generates real-time statistical metrics (e.g. top attacking IPs, targeted usernames).
Archived & Binary Logs (zgrep/journalctl)	Enables historical timeline analysis across log rotations and systemd service journals.